UMBC Parking Management System
Database Design Proposal

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March 11, 2026

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HW #1 Executive Summary

1.1 Problem Statement

Parking at a large commuter campus such as UMBC is inefficient, stressful, and difficult to enforce. Students frequently waste valuable time searching for open spaces. Visitors struggle to locate valid parking without clear guidance. Faculty and staff require access to restricted or premium areas. Meanwhile, enforcement is largely reactive and labor-intensive, relying on officers instead of automated validation mechanisms.

The core problem is the absence of a centralized, real-time database system that integrates parking permits, lot and spot management, vehicle data, reservations, sensor occupancy detection, and automated enforcement into a unified platform. Without such integration, parking operations lack efficiency, transparency, and scalability.

The proposed UMBC Parking Management System addresses this issue by implementing a PostgreSQL-backed database that manages parking as a real-time, data-driven ecosystem.

1.2 Parking Users

The system supports multiple categories of users:

- **Commuter Students** – Purchase student permits and park in designated student lots.
- **Faculty and Staff** – Hold faculty/staff permits and access restricted premium zones.
- **Visitors and Guest Lecturers** – Reserve temporary parking spaces online and pay per visit.

1.3 Scope of Implementation

This project will implement a PostgreSQL database system that includes:

1. **User and Vehicle Management**
    - Store user profiles (students, faculty, visitors).
    - Associate multiple vehicles with individual users.
    - Store license plate information for validation.
2. **Permit Management**
    - Define permit types (student, faculty, visitor).

- Assign permits to users.
- Enforce expiration dates.
- Restrict lot and zone access by permit type.

3. **Lot and Spot Management**
    - Define parking lots, zones, rows, and individual spots.
    - Track real-time occupancy per spot.
    - Associate spots with allowed permit types.
4. **Reservation System**
    - Allow visitors to reserve specific spots.
    - Prevent double-booking through constraints and transactions.
    - Lock spots upon confirmed payment.
5. **Sensor Integration**
    - Update occupancy when a ground sensor detects a vehicle.
    - Validate vehicle authorization against zone restrictions.
6. **Automated Ticketing**
    - Generate violations for unauthorized parking.
    - Store fine amounts and violation details.
    - Record notification events.
7. **Payments**
    - Record ticket payments.
    - Track outstanding balances.
8. **Reporting and Analytics**
    - Lot utilization rates.
    - Revenue reports.
    - Violation frequency reports.
    - Active and expired permit tracking.
    - Peak usage analysis.

9. **Concurrency Control**
    - Prevent race conditions during reservation.

This scope focuses specifically on the database design, implementation, constraints, indexing, realistic operations, and concurrency control. It does not include a full mobile or web application interface.

1.4 Concrete Use Cases

The database must support the following operations:

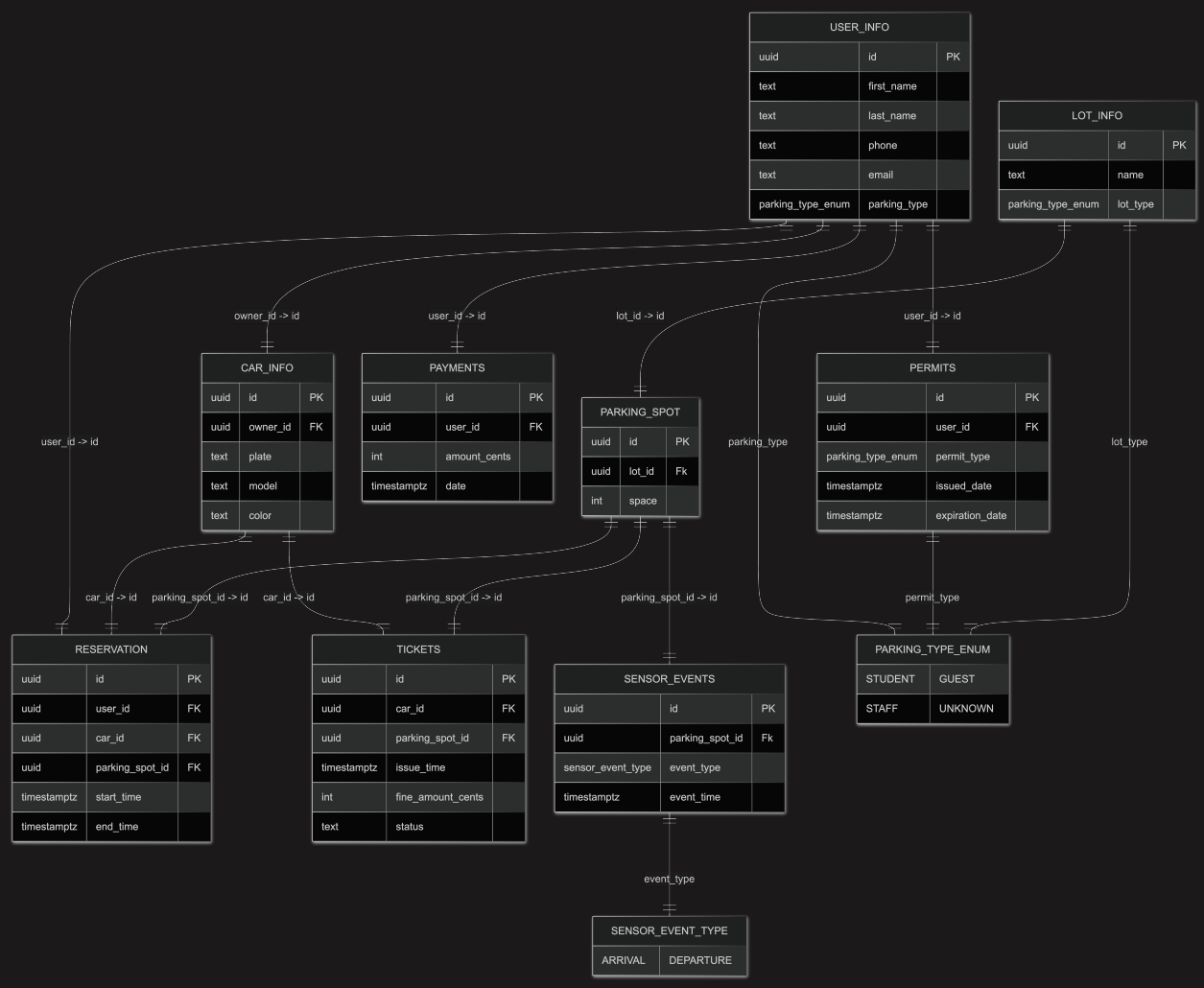
- Register a new user (student, faculty, or visitor).
- Add one or more vehicles to a user account.
- Purchase and assign a parking permit to a user.
- Define parking lots and individual parking spots.
- Associate permit types with allowed zones.
- Query real-time available parking spots in a specific lot.
- Create a visitor reservation for a specific date and time.
- Prevent double-booking of a parking spot.
- Update occupancy when a sensor detects a vehicle.
- Validate a vehicle's permit against the spot's allowed zone.
- Automatically generate a ticket for unauthorized parking.
- Record ticket payments and mark violations as resolved.
- Generate reports on lot occupancy over time.
- Generate revenue reports from permits, reservations, and fines.
- Detect expired permits and flag vehicles as invalid.
- Handle simultaneous reservation attempts without data corruption.

HW #2 Requirements & Conceptual Design

This section will more rigourously define the requirements, constraints, roles, as well as include an ER diagram.

2.1 ER Diagram

To see full image click on `er_diagram.png`



2.2 Requirements and Constraints

The following business rules define constraints for the UMBC Parking Management System. Each rule is labeled with how it is enforced: (A) through database table constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, or CHECK, or (B) through triggers, functions, or application logic.

Rule 1: Each user may register multiple vehicles, but each license plate must be unique across the system.

Enforcement: **(A)** A UNIQUE constraint is applied to the car_info.plate column.

Rule 2: Each parking spot within a lot must be uniquely identified by its lot and space number.

Enforcement: **(A)** A composite UNIQUE constraint is applied to (lot_id, space) in the parking_spot table.

Rule 3: A reservation must have a start time that occurs before its end time.

Enforcement: **(A)** A CHECK constraint ensures start_time <end_time.

Rule 4: A permit must expire after it is issued.

Enforcement: **(A)** A CHECK constraint ensures expiration_date >issued_date.

Rule 5: Monetary values for fines and payments must be positive.

Enforcement: **(A)** CHECK constraints ensure values such as fine_amount_cents >0 and amount_cents >0.

Rule 6: A parking spot cannot be reserved by two users for overlapping time intervals.

Enforcement: **(B)** Implemented using a PostgreSQL trigger that looks at the time range derived from start_time and end_time.

Rule 7: Only vehicles with a valid and unexpired permit may park in permit-restricted lots.

Enforcement: **(B)** Implemented using a trigger or application-level validation that checks the vehicle's permit type and expiration date.

Rule 8: Visitor reservations must be confirmed before the space is considered locked.

Enforcement: **(B)** Application logic ensures reservations transition from PENDING to CONFIRMED only after successful payment.

Rule 9: Sensor events must record an arrival or departure event with a timestamp.

Enforcement: **(A)** Implemented using the sensor_event_type enum and a NOT NULL timestamp constraint.

Rule 10: If a vehicle parks without authorization (no valid permit or reservation), the system must generate a parking violation ticket.

Enforcement: **(B)** Implemented using a trigger that listens for arrival events and inserts a record into the tickets table when authorization fails.

2.3 Concurrency Scenario

A concurrency scenario occurs when two users attempt to reserve the same parking spot at nearly the same time. For example, User A and User B both attempt to reserve parking spot S in Lot 3 from 9:00 AM to 12:00 PM.

Without proper concurrency control, both reservations could be inserted into the database, creating a double booking. The system must guarantee that only one reservation succeeds.

To enforce this requirement, the database uses a PostgreSQL trigger that runs BEFORE insertion into the reservations table, that prevents overlapping time ranges for the same parking spot. When the first reservation transaction commits successfully, any subsequent reservation that overlaps with that time interval will automatically fail.

This approach ensures correctness even when multiple transactions occur simultaneously.

2.4 Rationale

The design separates long-term policy information from transactional records in order to improve flexibility and maintainability. Permit classifications (such as student, faculty, or visitor) represent policy rules that determine which parking areas are accessible to different groups. By separating permit types from individual permit records, the system allows parking policies to evolve independently from historical permit data.

Reservations are modeled using explicit start and end timestamps rather than a single date field so that the system can support flexible time windows. This design enables features such as hourly reservations, overlapping time checks, and historical analysis of parking usage patterns.

Sensor events are stored as individual event records instead of directly updating a parking spot's status. This event-driven model preserves a complete history of arrivals and departures, which is valuable for analytics, auditing, and debugging sensor behavior. A trigger or background process can interpret these events to update occupancy status, validate permits, or generate tickets when violations occur.

Finally, the schema models relationships such as vehicles belonging to users and reservations referencing parking spots using foreign keys. These relationships ensure referential integrity and make it possible to query the system efficiently for operations such as finding available parking, verifying permits, or generating enforcement reports.

HW #3 Logical Design

3.1 Schema List

Here `parking_type`, `lot_type`, `permit_type` and `event_type` are all SQL enum types. I have defined them in `db/migrations/01_enums.sql`

```
Lot_Info(id, name, lot_type)

User_Info(id, first_name, last_name, phone, email, parking_type)

Car_Info(id, plate, model, color, FK -> User_Info(id))

Parking_Spot(id, space, FK -> Lot_Info(id))

Reservation(id, start_time, end_time,
            FK -> User_Info(id),
            FK -> Car_Info(id),
            FK -> Parking_Spot(id))

Sensor_Events(id, event_type, event_time,
            FK -> Parking_Spot(id))

Permits(id, permit_type, issued_date, expiration_date,
        FK -> User_Info(id))

Payments(id, amount_cents, date,
        FK -> User_Info(id))

Tickets(id, issue_time, fine_amount_cents, status,
        FK -> Car_Info(id),
        FK -> Parking_Spot(id))
```